Energy Systems – Side Worksheets (Slides 24–61 Companion)

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Worksheet 1: Plain-English Explainer – Fatigue & Recovery

Use this as notes. Then highlight the key words you want to remember.

• Fatigue = when your muscles cannot keep the same effort. It happens for different reasons depending on intensity and duration.

• Short, explosive efforts (under ~10s) mainly use the ATP–CP system. Fatigue occurs when creatine phosphate (CP) runs low.

• Efforts ~10s–2 min at high intensity rely more on anaerobic glycolysis. Fatigue occurs because lactate and H+ (acid) build up, making muscles feel a burning sensation.

• Longer, steady efforts (2+ min) rely more on the aerobic system. Fatigue is more about fuel supply (carbs/fats), dehydration, and body temperature rising.

• Recovery means your body removes waste (like H+), replaces CP, and brings heart rate/breathing back down.

• Quick recovery (minutes) = rest, breathing control, light movement. Full recovery (hours) = sleep, fluids, food.

Worksheet 2: Sort & Match – Causes of Fatigue

Draw arrows between the activity and the most likely main fatigue cause. Add one sentence explaining why.

Activities: (A) 8-second sprint (B) 45-second shuttle run (C) 5-minute steady jog (D) 2-minute skipping at high pace

Fatigue causes: (1) CP depletion (2) H+ (acid) accumulation (3) Fuel use/dehydration/heat (4) Mixed cause

Your arrows: _____

One-sentence explanations:

1. _____

2. _____

3. _____

4. _____

Worksheet 3: Recovery Planner – What Works & When

Tick which recovery strategies fit each situation and explain why.

Situations:

1) After 3 × 10s maximal sprints

2) After 3 × 45s high-intensity shuttles

3) After a 10-minute continuous run

Recovery options: light walk, deep breathing, fluids, cool-down stretches, carbohydrate snack, ice/heat, sleep

1) Chosen options: _____ Why? _____

2) Chosen options: _____ Why? _____

3) Chosen options: _____ Why? _____

Worksheet 4: Anaerobic vs Aerobic – Evidence Table

Fill in the table using observations from class (HR, breathing, feeling, performance).

Activity Pair: 30s fast feet (anaerobic) vs 2-min jog (aerobic)

HR after: _____ vs _____ Breathing: _____ vs _____ Feeling: _____ vs _____

Which energy system was dominant for each? _____

Exit Ticket (Choose One to Answer)

1) Why does a 10s sprint feel different to a 2-min jog?

2) Name one recovery method and when you would use it.

3) What causes the “burn” feeling in hard efforts?